

ECE 647: Lecture 28

Duality

Convert a convex program (Primal Problem) with a set of variables (Primal Variables) to another convex program (Dual Problem) with another set of variables (Dual Variables)

Under very general conditions, the two problems are equivalent

Benefit of studying the Dual Problem:

1. Reveal additional structure of the optimal solution (KKT Condition)
 - The dual variables often have interesting engineering interpretations ("Price")
 - The original problem may be decomposed into sub-problems when we convert to the dual
 - distributed systems
- 2 The constraint set of the dual problem is often simpler
 - easy to compute projection

The primal problem:

$$\begin{array}{ll} \min & f_0(\mathbf{x}) \\ \text{s.t.} & f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & h_i(\mathbf{x}) = 0, \quad i = 1, \dots, p \end{array} \quad \begin{array}{l} \lambda \geq 0 \\ v \in \mathbb{R} \end{array}$$

multipliers $\uparrow$

Lagrangian:

$$L(x, \lambda, v) = f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{i=1}^p v_i h_i(\mathbf{x})$$

can be viewed as

the penalty if

$$f_i(x) > 0$$

- all plus signs when we use standard form
- $\lambda_i \geq 0 \Rightarrow L$ is convex in $\mathbf{x}$

Minimize the Lagrangian over prime variables

$$g(\lambda, v) = \min_{\mathbf{x}} L(x, \lambda, v)$$

$$\mathbf{D} = \text{dom}(f_0) \cap \text{dom}(f_i) \cap \text{dom}(h_i)$$

- Note: The constraints are eliminated

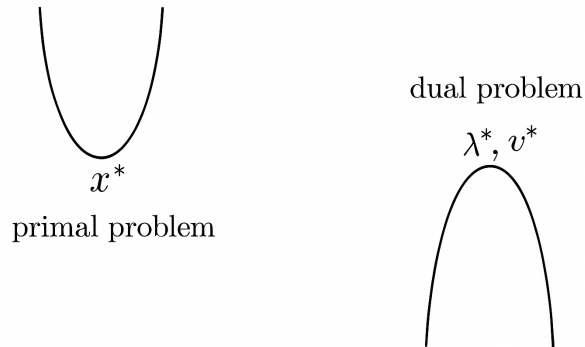
Dual Problem

$$\begin{array}{ll} \max & g(\lambda, v) \\ \text{s.t.} & \lambda \geq 0 \end{array}$$

Remarks

1. We may view the additional terms in $L(\mathbf{x}, \lambda, v)$ as a penalty term when $\mathbf{x}$ is not feasible (i.e. $\mathbf{x}$ does not satisfy the original constraints)
 - Not quite though $\Rightarrow$ They also disturb the value of $f_0(\cdot)$ within the feasible set!
2. When we compare $g(\lambda, v)$, we have an unconstrained minimization over $\mathbf{x}$
3. $g(\lambda, v)$ is a concave function of λ & v
 - It is a minimization of a set of Linear functions of (λ, v)

4 dual problem is also a convex problem



5 The constraint set of the dual problem is a quadrant set
 $\Rightarrow$ Projection becomes trivial

Dual vs Primal

Q) When/Why does solving the dual also solve the primal?

Easy to see that:

$$\begin{aligned} g(\lambda, v) &\leq L(x^*, \lambda, v) \leq f_0(\mathbf{x}) \quad \forall \lambda \geq 0 \\ &\Rightarrow \max_{\lambda \geq 0} g(\lambda, v) \leq f_0(\mathbf{x}^*) \end{aligned}$$

The optimal value of dual is always no greater than the optimal value of the primal.

$\Rightarrow$ weak duality

Under light conditions, they are in fact equal

$\Rightarrow$ strong duality

Consequence of Strong Duality

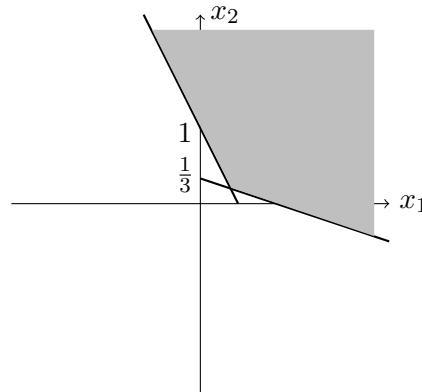
There exists (λ^*, v^*) such that

$$\begin{aligned} g(\lambda^*, v^*) &= L(\mathbf{x}^*, \lambda^*, v^*) = f_0(\mathbf{x}^*) \\ &\quad \downarrow \uparrow \\ \min_{\mathbf{x}} L(\mathbf{x}, \lambda^*, v^*) \end{aligned}$$

$\Rightarrow$ We can recover $\mathbf{x}^*$ from (λ^*, v^*) by minimizing $L(\mathbf{x}, \lambda^*, v^*)$ over $\mathbf{x}$.

EX

$$\begin{aligned} \min \quad & x_1^2 + 9x_2^2 \\ \text{s.t.} \quad & 2x_1 + x_2 \geq 1 & \lambda_1 \\ & x_1 + 3x_2 \geq 1 & \lambda_2 \end{aligned}$$



The Lagrangian is:

$$L(x_1, x_2, \lambda_1, \lambda_2) = x_1^2 + 9x_2^2 + \lambda_1(1 - 2x_1 - x_2) + \lambda_2(1 - x_1 - 3x_2)$$

$$\Rightarrow x_1^2 - (2\lambda_1 + \lambda_2)x_1 + 9x_2^2 - (2\lambda_1 + 3\lambda_2)x_2 + \lambda_1 + \lambda_2$$

Minimize the Lagrangian over x_1 & x_2 . Since there are no longer coupling constraints, in this case x_1 & x_2 can be optimized separately $\rightarrow$ "separable problems"

$$\begin{aligned}
\frac{\partial L}{\partial x_1} &= 2x_1 - (2\lambda_1 + \lambda_2) = 0 \\
\frac{\partial L}{\partial x_2} &= 18x_2 - (\lambda_1 + 3\lambda_2) = 0 \\
\Rightarrow x_1 &= \frac{2\lambda_1 + \lambda_2}{2} \\
x_2 &= \frac{\lambda_1 + 3\lambda_2}{18} \\
g(\lambda_1, \lambda_2) &= -\left(\frac{(2\lambda_1 + \lambda_2)^2}{4}\right) - \frac{(\lambda_1 + 3\lambda_2)^2}{36} + \lambda_1 + \lambda_2
\end{aligned}$$

The Dual Problem:

$$\begin{aligned}
\max \quad & g(\lambda_1, \lambda_2) \\
\text{s.t.} \quad & \lambda_1 \geq 0 \quad \lambda_2 \geq 0
\end{aligned}$$

To solve the dual problem:

$$\begin{aligned}
\frac{\partial g}{\partial \lambda_1} &= -\frac{2(2\lambda_1 + \lambda_2)}{4} \cdot 2 - \frac{2(\lambda_1 + 3\lambda_2)}{36} + 1 \\
&= -(2\lambda_1 + \lambda_2) - \frac{\lambda_1 + 3\lambda_2}{18} + 1 \\
&= -\frac{37}{18}\lambda_1 - \frac{7}{6}\lambda_2 + 1 \\
\\
\frac{\partial g}{\partial \lambda_2} &= -\frac{2(2\lambda_1 + \lambda_2)}{4} - \frac{2(\lambda_1 + 3\lambda_2)}{36} \cdot 3 + 1 \\
&= -\frac{2\lambda_1 + \lambda_2}{2} - \frac{\lambda_1 + 3\lambda_2}{6} + 1 \\
&= -\frac{7}{6}\lambda_1 - \lambda_2 + 1
\end{aligned}$$

Suppose all constraints are not binding. We can set $\frac{\partial g}{\partial \lambda_1} = 0$ & $\frac{\partial g}{\partial \lambda_2} = 0$. We have:

$$\begin{cases} \frac{37}{18}\lambda_1 + \frac{7}{6}\lambda_2 = 1 \\ \frac{49}{36}\lambda_1 + \frac{7}{6}\lambda_2 = \frac{7}{6} \end{cases}$$

$$\frac{25}{36}\lambda_1 = -\frac{1}{6} \Rightarrow \lambda_1 = -\frac{6}{25}$$

$$\lambda_2 = 1 - \frac{7}{6}\lambda_1 = \frac{32}{25}$$

However, this violates the constraint on $\lambda_1 \geq 0$.

Second guess: $\lambda_1 = 0$

$$\frac{\partial g}{\partial 2\lambda_2} = 0$$

$$\begin{cases} \lambda_1^* = 0 \\ \lambda_2^* = 1 \end{cases}$$

This is the optimized point for the dual problem.

- Let us verify if strong duality holds.
- Suppose strong-duality holds. Then, we can find the optimal primal solution by minimizing $L(\mathbf{x}, \lambda_1^*, \lambda_2^*)$:

Hence,

$$\begin{aligned} x_1^* &= \frac{2\lambda_1 + \lambda_2}{2} = \frac{1}{2} \\ x_2^* &= \frac{\lambda_1 + 3\lambda_2}{18} = \frac{1}{6} \end{aligned}$$

Note that this is feasible for the primal problem:

$$\begin{aligned} 2x_1^* + x_2^* &= \frac{7}{6} \geq 1 & \lambda_1^* &= 0 \\ x_1^* + 3x_2^* &= 1 & \lambda_2^* &> 0 \end{aligned}$$

complimentary slackness $\uparrow$

Finally,

$$\begin{aligned} x_1^* + 9x_2^* &= \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \\ g(\lambda_1^*, \lambda_2^*) &= -\frac{1}{4} - \frac{1}{4} + 1 = \frac{1}{2} \end{aligned}$$

The duality gap is indeed zero!